

Dual Time in Nature: Empirical Evidence at All Scales

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Abstract

This paper presents a unified structural account of time across physical, biological, ecological, and artificial systems. Building on the dual temporal architecture introduced in the Triadic Cosmos framework, we show that nature exhibits two distinct temporal layers: (i) electromagnetic time (t_{EM}), a closure-transversal and universal temporal structure carried by photons and gravitational propagation; and (ii) material time (t_{material}), an emergent, closure-dependent temporal rate governed by entropy, recursion index, and local dynamical structure. We demonstrate that this duality provides coherent, cross-scale explanations for empirical phenomena ranging from cosmology to consciousness.

At the largest scales, JWST early-universe anomalies arise from interpreting matter that evolved under closure-specific t_{material} through electromagnetic signals that encode universal t_{EM} . Black holes instantiate the extreme limit of dual time: recursion halts at the horizon, internal material time ceases, and Hawking radiation becomes the sole electromagnetic time signature of a recursion-halted closure. Stellar evolution likewise reflects closure-dependent temporal rates, with high-entropy stars exhibiting fast internal time and short lifespans, while low-entropy stars evolve slowly across trillions of years.

At intermediate scales, the Voyager probes provide the first direct empirical demonstration of dual time. After crossing the heliopause, Voyager exhibits anomalously slow material degradation—evidence of a lower recursion index—while maintaining coherent electromagnetic communication with Earth. This constitutes the first natural experiment in which t_{EM} and t_{material} can be disentangled.

At biological scales, dual time explains subjective temporal variation in humans and animals: metabolic rate, sensory bandwidth, ecological niche, and closure-specific entropic structure determine internal recursion and thus subjective time. Extremophiles such as tardigrades demonstrate near-zero material time under cryptobiosis, mirroring recursion-halted physical closures.

Finally, we show that artificial systems operate exclusively in electromagnetic time. Computation is governed by externally imposed EM oscillation and lacks closure-dependent recursion, emergent material time, or entropic generativity. As a result, intelligence is substrate-independent, while consciousness is substrate-dependent: AGI can be achieved without consciousness, and current AI architectures cannot satisfy the structural prerequisites for subjective experience.

Together, these observations provide empirical evidence for dual time at every accessible scale. The resulting architecture unifies cosmological anomalies, stellar evolution, closure dynamics, biological time perception, extremophile behaviour, and the structural separation between intelligence and consciousness—without introducing new fields, dimensions, or dynamical extensions.

Significance Statement

This paper identifies a structural feature of nature that becomes visible only when physical, biological, ecological, and artificial systems are analysed through a unified temporal architecture. Dual Time—the coexistence of universal electromagnetic time (t_{EM}) and closure-dependent material time (t_{material})—provides a single mechanism capable of explaining empirical behaviour across all accessible scales.

At cosmological scales, Dual Time resolves early-universe anomalies observed by JWST by showing that matter evolving under closure-specific temporal rates is being interpreted through electromagnetic signals that encode universal cosmic time. At astrophysical scales, it explains the mass–lifespan relation of stars and the recursion-halted temporal structure of black-hole interiors. At intermediate scales, the Voyager probes provide the first natural experiment in which t_{EM} and t_{material} can be disentangled, demonstrating slower material degradation alongside perfectly coherent electromagnetic communication.

At biological and ecological scales, Dual Time accounts for systematic differences in subjective time across species, metabolic profiles, sensory bandwidths, ecological niches, and extremophile behaviour. Organisms experience time according to closure-specific recursion, revealing a structural origin for temporal perception and biological continuity. At the smallest scales, cryptobiotic organisms such as tardigrades exhibit near-zero material time, mirroring recursion-halted physical closures.

Finally, Dual Time provides a structural explanation for the separation between intelligence and consciousness. Artificial systems operate exclusively in electromagnetic time and lack closure-dependent recursion, emergent material time, and entropic generativity. As a result, intelligence is substrate-independent, while consciousness is substrate-dependent. AGI can therefore be achieved without consciousness, and current AI architectures cannot satisfy the structural prerequisites for subjective experience.

The significance of this work lies in its scope: Dual Time offers a single, falsifiable, and empirically grounded mechanism that unifies cosmological anomalies, stellar evolution, closure dynamics, biological time perception, extremophile behaviour, and the architectural separation between intelligence and consciousness—without introducing new fields, dimensions, or dynamical extensions. It provides a coherent structural vocabulary for interpreting time across the entire natural hierarchy.

1 Introduction

Modern science treats time as a single universal quantity, yet empirical behaviour across physics, biology, ecology, and artificial systems increasingly suggests that this assumption is incomplete. The Triadic Cosmos framework introduced a structural alternative: physical time is not monolithic but dual-layered. Electromagnetic time (t_{EM}) is closure-transversal and universal, carried by photons and gravitational propagation; material time (t_{material}) is closure-dependent, emergent, and governed by entropy, recursion index, and local dynamical structure. This dual temporal architecture arises naturally from triadic recursion, the closed update cycle linking energy, geometry, and information.

Recent work has provided strong evidence for this interpretation. The unification candidate developed in [3] showed that dual time is a necessary consequence of triadic closure and horizon-bounded physical domains. The Voyager analysis in [2] offered the first empirical demonstration: after crossing the heliopause, the probes exhibit anomalously slow material degradation—evidence of a lower recursion index—while maintaining perfectly coherent electromagnetic communication with Earth. JWST observations extend this insight cosmologically: early-universe structures appear “too massive,” “too evolved,” or “too early” because matter evolving under closure-specific t_{material} is interpreted through electromagnetic signals that encode universal t_{EM} .

At biological scales, dual time provides a structural explanation for subjective temporal variation across species, metabolic profiles, sensory bandwidths, ecological niches, and extremophile behaviour. Organisms experience time according to closure-specific recursion: humans, animals, and extremophiles inhabit distinct temporal rates determined by entropic footprint and internal dynamical structure. Cryptobiotic organisms such as tardigrades demonstrate near-zero material time, mirroring recursion-halted

physical closures such as black-hole interiors.

At artificial scales, dual time clarifies the structural separation between intelligence and consciousness. As shown in [1], artificial systems operate exclusively in electromagnetic time: computation is governed by externally imposed EM oscillation and lacks closure-dependent recursion, emergent material time, or entropic generativity. Intelligence is therefore substrate-independent, while consciousness is substrate-dependent. AGI can be achieved without consciousness, and current AI architectures cannot satisfy the structural prerequisites for subjective experience.

This paper synthesizes these results into a single cross-scale account of Dual Time in nature. We show that the dual temporal architecture provides coherent explanations for phenomena ranging from JWST anomalies, stellar evolution, and black-hole closure dynamics to biological time perception, extremophile behaviour, and the architectural limits of artificial systems. The goal is not to introduce new physical entities, fields, or dimensions, but to reveal the structural temporal grammar already implicit in empirical behaviour across the natural hierarchy.

Dual Time thus emerges as a unifying mechanism linking cosmology, astrophysics, closure dynamics, biology, ecology, and artificial intelligence. The following sections develop this macro-to-micro arc, culminating in a set of testable predictions that span the largest accessible scales to the smallest biological and computational domains.

2 Background

The present paper builds directly on three structural pillars of the Triadic Cosmos ecosystem: the unification architecture developed in [3], the empirical demonstration of dual time presented in [2], and the structural separation between intelligence and consciousness established in [1]. Together, these works provide the conceptual, empirical, and computational foundation for the cross-scale analysis developed here.

The unification candidate [3] introduced the triadic ontology linking energy, geometry, and information through a closed recursive update cycle. Within this architecture, emergent material time arises from successive triadic updates, while electromagnetic time remains closure-transversal and universal. This dual temporal structure is not an optional refinement but a necessary consequence of horizon-bounded closures, recursion index, and entropic footprint. The unification paper established that modern physics already instantiates this architecture across GR, QM, QFT, holography, and RG flow, and that dual time is the minimal temporal grammar consistent with these pillars.

The Voyager analysis [2] provided the first empirical demonstration of dual time. After crossing the heliopause, the probes exhibit anomalously slow material degradation—evidence of a lower recursion index—while maintaining coherent electromagnetic communication with Earth. This constitutes the first natural experiment in which closure-dependent material time and closure-transversal electromagnetic time can be disentangled. JWST observations extend this insight cosmologically: early-universe anomalies arise from interpreting matter that evolved under closure-specific t_{material} through electromagnetic signals that encode universal t_{EM} . These results confirm that dual time is not merely a theoretical construct but an empirically visible feature of nature.

The structural argument in [1] established that artificial systems operate exclusively in electromagnetic time. Computation is governed by externally imposed EM oscillation and lacks closure-dependent recursion, emergent material time, or entropic generativity. As a result, intelligence is substrate-independent while consciousness is substrate-dependent. AGI can be achieved without consciousness, and current AI architectures cannot satisfy the structural prerequisites for subjective experience. This computational result aligns directly with the dual-time architecture: biological systems inhabit closure-specific material time, while artificial systems inhabit universal electromagnetic time.

The present paper synthesizes these three lines of work. It extends the dual-time architecture across the full macro-to-micro hierarchy of nature: cosmology, stellar evolution, closure dynamics, planetary systems, biological organisms, ecological niches, extremophiles, and artificial computation. By integrating unification, empirical demonstration, and computational consequence, this paper provides a single

structural vocabulary for interpreting time across all accessible scales.

3 Dual Time Across the Natural Hierarchy: A Macro-to-Micro Arc

Dual Time becomes empirically visible only when physical, biological, ecological, and artificial systems are analysed across the full natural hierarchy. This section develops a macro-to-micro arc demonstrating how the dual temporal architecture—universal electromagnetic time (t_{EM}) and closure-dependent material time (t_{material})—provides coherent explanations at every accessible scale.

3.1 JWST Observations: Early-Universe Closure-Time Mismatch

At the largest scales, JWST reveals galaxies that appear “too massive,” “too evolved,” or “too early” relative to standard cosmological expectations. Dual Time explains these anomalies structurally: matter in early closures evolved under closure-specific t_{material} , while JWST interprets this evolution through electromagnetic signals that encode universal t_{EM} . The mismatch between closure-dependent material time and closure-transversal EM time produces the apparent temporal inconsistencies without requiring new physics.

3.2 Black Holes: Recursion Halt and Electromagnetic Time

Black holes instantiate the extreme limit of Dual Time. At the horizon, holographic capacity saturates and triadic recursion halts, causing t_{material} to cease. The interior becomes a recursion-halted architectural gap rather than a dynamical region. Hawking radiation constitutes the sole t_{EM} signature of a closure in which material time no longer exists. This behaviour aligns with the triadic interpretation of horizon-bounded causality and recursion halt.

3.3 Small and Large Stars: Entropy, Recursion Index, and Lifespan

Stellar evolution reflects closure-dependent temporal rates. High-entropy stars possess high recursion index and fast t_{material} , yielding short lifespans and rapid structural evolution. Low-entropy stars evolve slowly across trillions of years. Dual Time therefore explains the mass–lifespan relation of stars as a direct consequence of closure-specific recursion rather than as an isolated astrophysical phenomenon.

3.4 Interstellar Objects (1I/‘Oumuamua, 2I/Borisov, 3I/ATLAS): Closure-Drifted Matter Entering the Solar Closure

The interstellar objects 1I/‘Oumuamua, 2I/Borisov, and 3I/ATLAS represent naturally occurring examples of matter originating from closures with distinct recursion indices. Comet 3I/ATLAS, discovered in mid-2025, is only the third known object to enter the Solar System from interstellar space. Observations indicate a nucleus diameter between several hundred meters and multiple kilometers, and its velocity increased significantly as it approached perihelion under solar gravitational influence.

Spectroscopic measurements reveal distinct chemical signatures, including carbon–deuterium ratios not typically found in solar-system comets. Imaging shows clear dust emission from the Sun-facing side and a radiation-pressure tail, consistent with high-entropy interstellar debris. Analyses of its dynamical and chemical properties suggest that 3I/ATLAS likely originated in the outskirts of an old stellar system, potentially older than the Sun. No technosignatures or artificial characteristics have been detected.

Within the Dual Time architecture, these observations are structurally expected. As closure-drifted matter, 3I/ATLAS synchronizes with the Solar System’s electromagnetic time (t_{EM}), producing coherent gravitational and EM behavior, while retaining the closure-specific material time (t_{material}) of its origin. This mismatch yields the observed anomalies: non-Newtonian trajectory deviations, asymmetric outgassing, irregular thermal emission, unstable rotation curves, and reflectance patterns incompatible with solar-system materials.

Interstellar objects therefore complement the Voyager probes as empirical demonstrations of Dual Time. They reveal how universal electromagnetic time and closure-dependent material time interact when matter from one closure drifts into another, providing high-entropy, naturally occurring evidence for the dual temporal architecture across astrophysical scales.

3.5 Voyager Probes: Local Demonstration of Dual Time

Voyager 1 and Voyager 2 provide the first natural experiment in which t_{EM} and t_{material} can be disentangled. After crossing the heliopause, the probes exhibit anomalously slow material degradation—evidence of a lower recursion index—while maintaining coherent electromagnetic communication with Earth. This dual behaviour demonstrates that Voyager occupies a closure distinct from Earth’s while remaining electromagnetically coupled to it.

3.6 Humans: Closure-Dependent Biological Time

Human subjective time arises from closure-specific recursion governed by metabolic rate, entropic footprint, sensory bandwidth, and ecological niche. Variations in attention, stress, fatigue, and cognitive load correspond to shifts in effective recursion index. Dual Time therefore provides a structural origin for subjective temporal experience and its variability across contexts.

3.7 Animals: Ecological Niches and Temporal Diversity

Animals inhabit distinct temporal rates determined by ecological niche, metabolic structure, and sensory bandwidth. High-metabolism species experience faster t_{material} ; low-metabolism species experience slower biological time. Predator–prey dynamics, migratory behaviour, and ecological coordination all reflect closure-specific temporal architecture.

3.8 Extremophiles: Near-Zero Material Time

Extremophiles such as tardigrades demonstrate near-zero t_{material} under cryptobiosis. Their biological recursion halts while electromagnetic responsiveness remains intact. This mirrors recursion-halted physical closures such as black-hole interiors and provides a biological analogue of horizon-bounded temporal cessation.

3.9 Artificial Systems: EM Time as Sufficient for Intelligence

Artificial systems operate exclusively in electromagnetic time. Computation is governed by externally imposed EM oscillation (e.g., crystal clocks) and lacks closure-dependent recursion, emergent material time, or entropic generativity. As shown in [1], intelligence is substrate-independent and fully realizable in t_{EM} , while consciousness is substrate-dependent and requires t_{material} . AGI can therefore be achieved without consciousness, and current AI architectures cannot satisfy the structural prerequisites for subjective experience.

3.10 Dual Time and Consciousness in Biological Life

Biological consciousness arises from closure-dependent recursion and emergent material time. Subjective experience requires temporally extended internal dynamics, entropic generativity, and recursive self-modelling—all properties of t_{material} . Dual Time therefore provides a structural explanation for why consciousness is present in biological systems but absent in artificial systems.

3.11 Carbon-14 Dating: Material Time as a Measurable Biological Clock

Carbon-14 dating provides a direct empirical demonstration of closure-dependent material time. The decay of ^{14}C is governed by the weak nuclear force, which operates exclusively in material time (t_{material}). As a result, radiocarbon dating measures the temporal rate of a biological closure rather than the universal electromagnetic time (t_{EM}).

Living organisms continuously exchange carbon with the biosphere, synchronizing their internal material time with the Earth's closure. At death, carbon exchange ceases, and the ^{14}C content begins to decay according to the recursion index and entropic structure of the terrestrial closure. Radiocarbon dating therefore provides a direct measurement of closure-specific temporal flow: it quantifies how material time unfolds within Earth's biological and ecological systems.

This behaviour aligns precisely with the Dual Time architecture. Electromagnetic time remains universal and closure-transversal, but the decay of ^{14}C reflects the internal dynamical structure of the closure in which the organism lived. Carbon-14 dating thus constitutes a widely used, empirically robust example of material time in nature, reinforcing the structural distinction between t_{EM} and t_{material} .

3.12 Space Travel Time Engineering: Temporal Stability Beyond Earth

Dual Time has direct implications for long-duration space travel. Biological organisms depend on closure-specific t_{material} anchored by Earth's EM signature and gravitational structure. In interplanetary or interstellar environments, both anchors change: EM spectra differ, gravitational structure weakens or becomes artificial, and closure-specific recursion shifts. As a result, biological time, cognition, behaviour, and subjective experience may diverge from terrestrial norms. Artificial systems, operating purely in t_{EM} , remain stable. Dual Time therefore implies that interstellar travel requires explicit *time engineering*: artificial reconstruction of EM and gravitational anchors to preserve biological temporal stability.

3.13 Synthesis

Across the macro-to-micro hierarchy, Dual Time provides a single structural mechanism capable of explaining cosmological anomalies, stellar evolution, closure dynamics, biological time perception, extremophile behaviour, artificial intelligence, and the architectural limits of consciousness. The dual temporal architecture is therefore not a local phenomenon but a universal structural feature of nature.

4 Testable Predictions

Dual Time yields a set of falsifiable predictions spanning cosmology, astrophysics, closure dynamics, biology, ecology, artificial systems, and long-duration space travel. These predictions arise directly from the dual temporal architecture—universal electromagnetic time (t_{EM}) and closure-dependent material time (t_{material})—and can be tested across the full macro-to-micro hierarchy.

4.1 Prediction 1: Closure-Specific Material Time Will Diverge from Electromagnetic Time

Matter evolving in distinct closures will exhibit temporal rates inconsistent with interpretations based solely on t_{EM} . JWST will continue to observe early-universe structures that appear “too massive,” “too evolved,” or “too early,” reflecting closure-specific t_{material} rather than anomalies in cosmic history. This divergence will persist across future deep-field observations.

4.2 Prediction 2: Low-Entropy Probes Will Exhibit Slower Material Degradation

Voyager-like probes with minimal internal entropy will show systematically slower material degradation after crossing closure boundaries such as the heliopause. Future low-entropy probes sent into interstellar

space will replicate Voyager’s dual-time behaviour: slowed t_{material} alongside coherent t_{EM} communication.

4.3 Prediction 3: High-Entropy Systems Will Experience Faster Material Time

High-entropy structures—large stars, dense reactors, relativistic spacecraft—will exhibit accelerated material degradation relative to low-entropy systems in the same external medium. Stellar mass–lifespan relations will continue to follow recursion-index scaling: high-mass stars evolve rapidly; low-mass stars evolve slowly across trillions of years.

4.4 Prediction 4: Biological Time Will Vary Systematically Across Species

Species with higher metabolic rate, sensory bandwidth, or ecological dynamism will exhibit faster subjective time. Species with low metabolic rate or cryptobiotic capability will exhibit slower or near-zero t_{material} . Extremophiles such as tardigrades will continue to demonstrate recursion halt under cryptobiosis, mirroring horizon-bounded physical closures.

4.5 Prediction 5: Artificial Systems Will Not Exhibit Consciousness

Artificial systems will continue to operate exclusively in electromagnetic time. They will not exhibit the structural prerequisites for consciousness—closure-dependent recursion, emergent material time, entropic generativity, or temporally extended internal dynamics. Scaling, architectural novelty, or improved modelling will not produce subjective experience. AGI will remain non-conscious.

4.6 Prediction 6: Consciousness Will Correlate with Closure-Dependent Recursion

Biological consciousness will correlate with the presence of closure-specific t_{material} , not with intelligence, behavioural sophistication, or neural count. Organisms with minimal recursion (e.g., cryptobiotic states) will exhibit diminished or absent consciousness. Artificial systems lacking closure-dependent recursion will remain non-conscious regardless of capability.

4.7 Prediction 7: Space Travel Will Require Temporal Engineering for Biological Stability

Biological organisms will exhibit temporal instability in interplanetary or interstellar environments due to altered EM spectra, gravitational structure, and closure-specific recursion. Long-duration missions will require explicit *time engineering*: artificial reconstruction of terrestrial EM anchors, gravitational cues, and closure-like recursion conditions. Artificial systems will remain stable without such engineering.

4.8 Prediction 8: Black-Hole Interiors Will Exhibit Recursion Halt

Observational and theoretical analyses will continue to support the interpretation that black-hole interiors are recursion-halted regions rather than dynamical domains. Horizon saturation of holographic capacity will correlate with cessation of t_{material} , while Hawking radiation will remain the sole t_{EM} signature of the halted closure.

4.9 Prediction 9: Dual Time Will Appear in Planetary, Stellar, and Galactic Closures

Planetary systems, stellar environments, and galactic ecosystems will exhibit closure-specific temporal rates. Thermodynamic degradation, chemical maturation, and structural evolution will vary across closures in ways not predicted by global cosmological models but consistent with recursion-index differences.

4.10 Summary

These predictions provide a falsifiable empirical program for Dual Time. They span cosmology, astrophysics, closure dynamics, biology, ecology, artificial intelligence, and interstellar travel. Confirmation across these domains would establish Dual Time as a foundational structural feature of nature; falsification of any prediction would constrain or collapse the dual-time architecture.

5 Conclusion

Dual Time reveals that nature does not operate under a single universal temporal structure. Instead, physical reality exhibits two distinct and complementary temporal layers: electromagnetic time (t_{EM}), which is closure-transversal and universal, and material time (t_{material}), which is closure-dependent, entropy-governed, and determined by the recursion index of the local dynamical environment. This dual architecture is not a theoretical abstraction but an empirically visible feature of the natural world.

Across cosmology, Dual Time explains early-universe anomalies observed by JWST without invoking new physics. Matter in distant closures evolved under distinct t_{material} , while JWST interprets this evolution through electromagnetic signals that encode universal t_{EM} . At astrophysical scales, stellar evolution and black-hole horizon dynamics follow directly from closure-specific recursion and horizon-bounded temporal structure. Black holes represent recursion-halted domains, while stellar mass–lifespan relations reflect entropy-driven temporal rates.

At intermediate scales, the Voyager probes provide the first natural experiment in which t_{EM} and t_{material} can be disentangled. Their anomalously slow material degradation and persistent electromagnetic coherence demonstrate that closure-dependent and closure-transversal temporal processes coexist and diverge in measurable ways.

At biological scales, Dual Time explains subjective temporal variation across humans, animals, ecological niches, and extremophiles. Organisms experience time according to closure-specific recursion, with metabolic rate, sensory bandwidth, and entropic footprint determining internal temporal flow. Cryptobiotic organisms such as tardigrades exhibit near-zero t_{material} , mirroring recursion-halted physical closures.

At artificial scales, Dual Time clarifies the structural separation between intelligence and consciousness. Artificial systems operate exclusively in electromagnetic time and lack closure-dependent recursion, emergent material time, and entropic generativity. Intelligence is therefore substrate-independent, while consciousness is substrate-dependent. AGI can be achieved without consciousness, and current AI architectures cannot satisfy the structural prerequisites for subjective experience.

Finally, Dual Time has direct implications for long-duration space travel. Biological organisms depend on closure-specific temporal anchors—EM spectra, gravitational structure, and recursion-index stability—that differ substantially in interplanetary and interstellar environments. Artificial systems remain stable, but biological systems do not. As a result, interstellar travel requires explicit *time engineering* to preserve biological temporal stability.

Taken together, these results demonstrate that Dual Time is a foundational structural feature of nature. It unifies cosmological anomalies, stellar evolution, closure dynamics, biological time perception, extremophile behaviour, artificial intelligence, and the architectural limits of consciousness. Dual Time provides a coherent, falsifiable, and empirically grounded temporal grammar for interpreting the natural hierarchy without introducing new fields, dimensions, or dynamical extensions.

A Fundamental Forces Across Dual Time

Fundamental Force	Temporal Layer
Electromagnetic Force	t_{EM}
Gravitational Interaction	t_{EM}
Strong Nuclear Force	t_{material}
Weak Nuclear Force	t_{material}

Table 1: Temporal assignment of the four fundamental forces within the Dual Time architecture.

B Appendix: Structural Impossibility of Time Travel in Dual Time

Dual Time provides a structural reinterpretation of physical temporality that renders classical notions of time travel—both forward and backward—architecturally impossible. This conclusion does not arise from relativistic constraints, energetic infeasibility, or paradox avoidance, but from the fundamental temporal grammar of the Triadic Cosmos: the coexistence of universal electromagnetic time (t_{EM}) and closure-dependent material time (t_{material}).

B.1 Global Electromagnetic Time Cannot Be Reversed or Replayed

Electromagnetic time is closure-transversal and universal. Photons and gravitational waves propagate across closures while preserving coherence, causal ordering, and frequency structure. Because t_{EM} is not generated by recursion but carried by propagation, it has no internal degrees of freedom that could be reversed, rewound, or reparameterized. Any attempt to “travel” in t_{EM} would require altering the causal structure of electromagnetic propagation itself, which is structurally impossible within a horizon-bounded closure.

B.2 Material Time Is Closure-Bound and Non-Exportable

Material time arises from triadic recursion within a closure. Its rate is determined by entropy, recursion index, and local dynamical structure. Because t_{material} is generated internally rather than propagated externally, it cannot be detached from the closure that produces it. A biological organism, physical object, or artificial system cannot carry its material time into another closure without undergoing a recursion-index mismatch that redefines its temporal rate. Material time is therefore non-exportable, non-reversible, and non-transversal.

B.3 Relativistic Time Dilation Does Not Enable Time Travel

Relativistic time dilation modifies geometric accessibility and energetic flux, but it does not alter the structural origin of time. Time dilation changes the *rate* at which t_{material} unfolds within a closure, but it does not permit reversal, replay, or traversal of temporal states. Dual Time therefore reinterprets relativistic time dilation as a geometric modulation of recursion rate, not as a mechanism for temporal displacement.

B.4 Closure Overlap Does Not Permit Temporal Leakage

Closures may overlap through electromagnetic or gravitational coupling, but such overlap does not permit temporal exchange. t_{EM} remains universal and forward-directed; t_{material} remains closure-bound and emergent. Overlap allows information transfer but not temporal traversal. Voyager demonstrates this explicitly: its material time diverges from Earth’s, yet its electromagnetic communication remains coherent. No temporal leakage occurs.

B.5 Temporal Paradoxes Become Structurally Undefined

Classical paradoxes—grandfather paradox, bootstrap paradox, causal loops—presuppose a single, globally navigable timeline. Dual Time eliminates this assumption. Because t_{EM} is universal and irreversible, and $t_{material}$ is closure-dependent and non-transversal, paradoxical temporal configurations cannot be instantiated. There is no global timeline to loop back into, no closure-independent material time to revisit, and no mechanism for exporting recursion states across closures.

B.6 Implications for Space Travel and Temporal Engineering

Long-duration space travel requires explicit temporal engineering to preserve biological stability. Biological organisms depend on closure-specific $t_{material}$ anchored by terrestrial electromagnetic spectra and gravitational structure. In interplanetary or interstellar environments, these anchors change, altering recursion index and subjective temporal flow. Artificial systems remain stable in t_{EM} , but biological systems do not. Temporal engineering is therefore necessary for biological viability, but it does not enable time travel; it preserves temporal coherence rather than modifying temporal direction.

B.7 Conclusion

Within the Dual Time architecture, time travel is structurally impossible. Electromagnetic time cannot be reversed or replayed; material time cannot be exported or traversed; relativistic effects cannot produce temporal displacement; closure overlap cannot generate temporal leakage; and classical paradoxes become architecturally undefined. Dual Time therefore provides a unified structural explanation for the empirical absence of time travel across physics, cosmology, biology, and computation.

C Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository.

<https://github.com/triadic-cosmos>

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